

MERN STACK

E-Commerce Web App

Roman Urdu Mein — Zero Se Hero Tak

CHAPTER 1

BEGINNER

THEORY + CODE

ROMAN URDU

CHAPTER 1

Introduction & Complete Setup

MERN kya hai? Tools install karo. Pehla project shuru karo.

Is Chapter Mein Aap Seekhenge:

- MERN Stack ka matlab samajhna
- E-Commerce project ka overview dekhna
- Node.js, npm, VS Code install karna
- Project folder aur package.json banana
- Pehla Node.js script likhna aur chalana
- Dependencies ka concept samajhna

■ Prerequisites

Koi programming knowledge zaroor nahi! Sirf computer chalana aana chahiye. Hum bilkul zero se shuru karenge — jaise ek baccha chalna seekhta hai! ■

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SECTION 1 — MERN Stack Kya Hai?

1.0 — MERN Stack Kya Hai?

Socho aap ek ghar bana rahe ho. Alag alag cheezein chahiyein — eentein, cement, darwaze. Usi tarah website banane ke liye bhi alag alag technologies chahiyein. MERN Stack in technologies ka ek kaamiyaab group hai.

MERN ek acronym hai — har letter ek technology ka naam hai:

M	MongoDB Database	Data store karta hai. Jaise almari jisme cheezein rakhte hain.
E	Express.js Backend Framework	Server ka kaam asaan karta hai. Helpful assistant ki tarah.
R	React.js Frontend Library	Website ka chehra — jo user screen pe dekhta hai.
N	Node.js JS Runtime	JavaScript ko computer pe chalata hai. JS ka ghar.

■ TIP

MERN ka sab se bada faida: Sirf EK language — JavaScript — se poori website banta hai! Frontend (React), Backend (Node+Express), Database (MongoDB) — teeno mein JS use hoti hai!

1.1 — Yeh Sab Milke Kaise Kaam Karte Hain?

Amazon pe 'shoes' search karte ho — parda ke peeche kya hota hai?

Step	Kaun?	Kya Hota Hai?
1	React	Tum search box mein shoes likhte ho
2	Express	React, server ko request bhejta hai: mujhe shoes chahiyein

3	MongoDB	Express database se shoes ka data maangta hai
4	MongoDB	Data wapas Express ko deta hai
5	Express	Data React ko forward karta hai
6	React	Screen pe shoes ki list dikha deta hai — kaam ho gaya!

SECTION 2 — Humara E-Commerce Project

2.0 — Hum Kya Banayenge? (ShopEase)

Is poore course mein hum **ShopEase** naam ka ek E-Commerce website banayenge — bilkul Amazon ya Daraz ka chhota aur real version!

■	Home Page	Products list, categories, search bar
■	Product Detail	Photo, description, price, reviews
■	Shopping Cart	Products add karo, quantity, total price
■	User Auth	Register, Login, Profile — secure JWT se
■	Checkout	Address, payment, order place karna
■	My Orders	Order history, status tracking
■	Admin Panel	Products/orders/users manage karna

2.1 — Technologies Jo Hum Use Kareng

Technology	Version	Kaam Kya Hai?
Node.js	v20+	JavaScript server chalane ke liye
Express.js	v4.18+	Backend API banane ke liye
MongoDB	v7+	Database — data store karna
Mongoose	v8+	MongoDB ko asaan banana (ODM)
React.js	v18+	Frontend — user interface
Redux Toolkit	v2+	App ka state manage karna
JWT	v9+	User authentication
bcrypt	v5+	Password encrypt karna
dotenv	v16+	Secret keys protect karna

SECTION 3 — Tools Install Karna

3.0 — Kya Kya Install Karna Hai?

Jaise carpenter ko pehle auzaar chahiye, waise humen bhi tools chahiye. Aur khushkhabri: yeh sab bilkul **FREE** hain!

3.1 — Node.js Install Karna

Node.js kya hai? Normally JavaScript sirf browser mein chalta tha. Node.js ne magic kiya — ab JavaScript **computer pe bhi chal sakti hai** bina browser ke!

■ TIP

Node.js ko ek translator socho. Tumne JS likhi -> Node.js ne computer ki language mein translate kiya -> computer ne kaam kar diya!

01	nodejs.org pe jao	Browser mein nodejs.org likho aur Enter dabao
02	LTS version download karo	LTS (Long Term Support) button — stable version
03	Installer chalao	Downloaded file pe double-click -> Next -> Next -> Install
04	Terminal kholna	Windows: Start -> cmd Mac: Spotlight -> terminal
05	Version check karo	node --version likho — v20.x.x aana chahiye

3.2 — Terminal Mein Check Karo

```
# Node.js version check  
  
node --version  
  
# Output: v20.11.0  
  
# npm version check  
  
npm --version  
  
# Output: 10.2.4
```

■ SAHI KIYA

Version number aa gaya? MUBARAK HO! Install ho gaya. Agar 'command not found' aaye toh computer restart karo ya dobara install karo.

3.3 — VS Code Install Karna

VS Code ek super-smart code editor hai — galtiyan pakad le, code rangeen kare, typing mein help kare!

1. code.visualstudio.com pe jao
2. Apna OS (Windows/Mac/Linux) wala button click karo
3. Installer chalao — Next > Next > Install
4. VS Code kholne ke baad dark theme choose karo!

3.4 — VS Code Extensions (Zaroor Install Karo!)

Extension	Kya Kaam Karta Hai?
Prettier	Code automatically format — hamesha clean dikhta hai
ESLint	Galtiyan real-time mein batata hai
ES7+ React Snippets	React ke shortcuts — jaldi code likhne mein help
Thunder Client	API test karne ke liye (Postman ka alternative)
MongoDB for VS Code	Database directly VS Code se dekho
Auto Rename Tag	HTML/JSX tags automatically rename hote hain
GitLens	Git changes track karna bohat asaan

SECTION 4 — Project Setup Shuru Karo

4.0 — Project Folder Banana

Jaise ghar banane se pehle jagah choose karte hain, waise coding mein pehle **project folder** banate hain. Saara code is folder mein rahega.

```
# Desktop pe jao

cd Desktop

# ShopEase folder banao

mkdir shopease

# Folder ke andar jao

cd shopease

# VS Code mein kholna

code .
```

■ TIP

mkdir = folder banana | cd = folder ke andar jaana | code . = VS Code mein kholna

4.1 — package.json Banana (npm init)

package.json project ka **ID card** hai — naam, version, packages sab yahan hote hain.

```
npm init -y

# -y matlab: sab defaults accept karo
```

```
{
  "name": "shopease",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo Error && exit 1"
  },
  "keywords": [],
```

```
"author": "",  
"license": "ISC"  
}
```

4.2 — Har Line Ka Matlab

Field	Value	Matlab
"name"	"shopease"	Project naam — lowercase, no spaces
"version"	"1.0.0"	1.0.0 = pehli stable release
"main"	"index.js"	Entry point — yahan se program start hoga
"scripts"	{...}	Command shortcuts — hum baad mein add karenge
"author"	"..."	Tumhara naam — apna naam zaroor likho!
"license"	"ISC"	ISC = free use ki permission

SECTION 5 — Pehla Node.js Script Likhna

5.0 — server.js File Banana

VS Code mein shopease folder mein nayi file banao: **server.js**

Left sidebar -> folder pe cursor -> + icon -> naam: server.js -> Enter

5.1 — Pehla Program

```
// Yeh hamara pehla Node.js program hai!  
  
// Double slash = comment – computer ignore karta hai  
  
console.log('Hello World!');  
  
console.log('Mera naam ShopEase hai!');  
  
console.log('Main ek E-Commerce website hoon!');
```

5.2 — Har Line Ka Detail Mein Matlab

// comment	Double slash = comment. Computer is line ko skip karta hai. Developer ke liye notes.
console	Built-in JS object. console ka matlab hai terminal/screen.
.log()	log ek function hai — andar jo likho usse screen pe print karo.
'Hello World!'	Yeh string hai — text. Quotes ke andar likhte hain. Single ya double dono chalte.
semicolon ;	Line ke aakhir mein — jaise sentence ke baad full stop. Statement khatam.

5.3 — Program Chalana

```
node server.js  
  
# Output:  
# Hello World!  
# Mera naam ShopEase hai!  
# Main ek E-Commerce website hoon!
```

■ SAHI KIYA

Output aa gaya? MUBARAK HO! Tumne pehla Node.js program chalaya! Yeh wahi pehla qadam hai jo har developer uthata hai!

5.4 — Variables

Variable ek **dabba** hai jisme value rakhte hain aur naam dete hain.

```
const shopName = 'ShopEase'; // const: value kabhi nahi badlegi
const taxRate = 0.05;

let cartItems = 0;           // let: value baad mein badal sakti hai
cartItems = 3;               // badal gaya!

console.log('Shop:', shopName); // Shop: ShopEase
console.log('Tax:', taxRate);    // Tax: 0.05
console.log('Cart:', cartItems); // Cart: 3
```

5.5 — JavaScript Data Types

Data Type	Example	Kab Use Hoga?
String	'ShopEase' ya "Hello"	Text ke liye — product naam, user naam
Number	42 ya 3.14 ya -10	Sankhya — price, quantity, age
Boolean	true ya false	Haan/Nahi — logged in hai ya nahi?
Array	['shoes', 'shirt', 'bag']	List ke liye — products, cart items
Object	{ name: 'Shoes', price: 100 }	Related data — ek product ki poori info
null	null	Jaanbujh kar khaali — koi value nahi
undefined	undefined	Variable banaya, value nahi di — auto aata hai

SECTION 6 — Dependencies Ka Concept

6.0 — npm Packages Kya Hote Hain?

Roti banana ho toh aata bazar se laate ho — khud nahi ugate. Coding mein bhi dusron ke banaye hue **packages** npm se download karte hain — mostly FREE! 1.3 million+ packages available hain!

6.1 — Package Install Karna

```
npm install express

# ya short form:

npm i express

# Package.json update ho jaata hai:

# "dependencies": { "express": "^4.18.2" }
```

6.2 — dependencies vs devDependencies

Type	Command	Kab?	Examples
dependencies	npm install X	Production + Dev — zaroori	express, mongoose
devDependencies	npm install X --save-dev	Sirf Dev mein	nodemon, jest, eslint

6.3 — node_modules & .gitignore

Package install hone ke baad npm ek **node_modules** folder banata hai jisme package ka poora code aata hai — yeh 100MB+ bhi ho sakta hai!

■ YAAD RAHO

node_modules kabhi GitHub pe mat bhejo — bahut bada hai! Koi bhi npm install karke khud download kar sakta hai.

```
# .gitignore file mein likho:

node_modules/

.env

.DS_Store

dist/
```

6.4 — nodemon — Auto Restart Tool

nodemon automatically server restart karta hai jab bhi code change karo. Bina nodemon ke har baar manually restart karna padta — bohat thakawat!

```
npm install nodemon --save-dev

# package.json scripts update karo:
{
  "scripts": {
    "start": "node server.js",
    "dev": "nodemon server.js"
  }
}

# Development mein hamesha yeh use karo:
npm run dev

# Code save karo -> server auto-restart! Magic!
```

■ TIP

npm run dev ab se best friend hai! Hamesha development mein yeh use karo.

SECTION 7 — Summary, Practice & Next Steps

7.0 — Is Chapter Ka Recap

■ MERN Stack	MongoDB + Express + React + Node — ek poori website ka combo
■ Node.js	JavaScript ko computer pe chalane ka engine — browser nahi chahiye
■ npm	Package market — 1.3M+ tayyar tools FREE mein
■ VS Code	Super-smart code editor — galtiyan pakad le, code rangeen kare
■ package.json	Project ka ID card — naam, version, dependencies sab yahan
■ console.log()	Screen pe print karna — developer ka best friend
■ Variables	const aur let — values store karne ke dabbe
■ Data Types	String, Number, Boolean, Array, Object — alag tarah ki values
■ node_modules	Downloaded packages ka ghar — kabhi GitHub pe mat bhejo
■ .gitignore	Secrets aur bari files ko Git se door rakhna
■ nodemon	Auto-restart tool — development mein time bachata hai

7.1 — Practice Exercises

Sirf padhne se coding nahi aati — karna zaroori hai!

■ Easy Level

1. shopease folder banao aur npm init -y chalao
2. server.js mein apna naam aur shehar print karo
3. 3 variables: apna naam, umar, favourite color
4. .gitignore file banao

■ Medium Level

1. nodemon install karo aur npm run dev se chalao

2. package.json mein description field fill karo
3. Ek array banao jisme 5 products ke naam hon

■ Challenge Level

1. JS object banao: product ki पूरी info (naam, price, category, inStock)
2. Har property alag alag console.log se print karo
3. Price pe 5% tax lagao — final price calculate aur print karo

7.2 — Next Chapter Preview

Express.js aur HTTP — Pehla Web Server Banana

- Express.js kya hai aur kyun zaroori hai
- HTTP requests aur responses ka concept
- GET, POST, PUT, DELETE methods ka matlab
- Pehla web server — browser se open karna
- Routes banana — alag URLs handle karna
- JSON data send aur receive karna
- Middleware ka concept — ek bohat powerful idea!

Keep Going — Tum Kar Sakte Ho! ■

Aaj tumne bohat kuch seekha — MERN ka matlab, tools install, pehla program, aur dependencies. Yeh foundations hain jis pe पूरी building banti hai. Agar kuch samajh na aaye toh dobara padho, practice karo. **Persistence hi success hai!**

Chapter 2 mein milte hain!